RACE
Regular Analysis Through Continuous Exercise

Electromagnetic Waves and Wave Optics : RACE-01

1. Arrange the following electromagnetic radiations per quantum in the order of increasing energy:

A : Blue light B : Yellow light C : X–ray D : Radio wave

(A) B, A, D, C (B) D, B, A, C (C) A, B, D, C (D) C, A, B, D

PWO210

2. The energy associated with electric field is (U_E) and with magnetic field is (U_B) for an electromagnetic wave in free space. Then :

(A) $U_E = \frac{U_B}{2}$ (B) $U_E < U_B$ (C) $U_E = U_B$ (D) $U_E > U_B$

PWO211

3. A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive x-direction. At a particular point in space and time, $\vec{E} = 6.3\hat{j}V/m$. The corresponding magnetic field $\vec{B}$, at that point will be:

(A) $18.9 \times 10^{-8}\ \hat{k}T$ (B) $6.3 \times 10^{-8}\ \hat{k}T$ (C) $2.1 \times 10^{-8}\ \hat{k}T$ (D) $18.9 \times 10^{8}\ \hat{k}T$

PWO212

4. A 27 mW laser beam has a cross-sectional area of 10 mm^2. The magnitude of the maximum electric field in this electromagnetic wave is given by

[Given permittivity of space $\in_0 = 9 \times 10^{-12}$ SI units, Speed of light $c = 3 \times 10^8\ m/s$]

(A) 1 kV/m (B) 2 kV/m (C) 1.4 kV/m (D) 0.7 kV/m

PWO213

5. An electromagnetic wave is represented by the electric field $\vec{E} = E_0\hat{n}\sin[\omega t + (6y - 8z)]$. Taking unit vectors in x, y and z directions to be $\hat{\imath}, \hat{\jmath}, \hat{k}$, the direction of propogation $\hat{s}$, is :

(A) $\hat{s} = \frac{4\hat{j}-3\hat{k}}{5}$ (B) $\hat{s} = \frac{3\hat{i}-4\hat{j}}{5}$ (C) $\hat{s} = \left(\frac{-3\hat{j}+4\hat{k}}{5}\right)$ (D) $\hat{s} = \frac{-4\hat{k}+3\hat{j}}{5}$

PWO214

6. A plane electromagnetic wave, has frequency of $2.0 \times 10^{10}\ Hz$ and its energy density is $1.02 \times 10^{-8}\ J/m^3$ in vacuum. The amplitude of the magnetic field of the wave is close to

($\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \frac{Nm^2}{C^2}$ and speed of light = $3 \times 10^8\ ms^{-1}$)

(A) 180 nT (B) 160 nT (C) 150 nT (D) 190 nT

PWO215

7. Suppose that intensity of a laser is $\left(\frac{315}{\pi}\right) W/m^2$. The rms electric field, in units of V/m associated with this source is close to the nearest integer is

($\in_0 = 8.86 \times 10^{-12}\ C^2\ Nm^{-2}$; $c = 3 \times 10^8\ ms^{-1}$)

PWO216

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8. The electric field in an electromagnetic wave is given by $E = 56.5 \sin \omega (t - x/c) NC^{-1}$. Find the intensity of the wave if it is propagating along x-axis in the free space.
(Given $\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)
(A) 5.65 Wm^{-2} (B) 4.24 Wm^{-2} (C) 1.9×10^{-7} Wm^{-2} (D) 56.5 Wm^{-2}

PWO217

9. The electric fields of two plane electromagnetic plane waves in vacuum are given by
$\vec{E}_1 = E_0 \hat{j} \cos(\omega t - kx)$ and $\vec{E}_2 = E_0 \hat{k} \cos(\omega t - ky)$
At $t = 0$, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced by the particle is :
(A) $E_0 q(-0.8\hat{i} + \hat{j} + \hat{k})$ (B) $E_0 q(0.8\hat{i} - \hat{j} + 0.4\hat{k})$
(C) $E_0 q(0.8\hat{i} + \hat{j} + 0.2\hat{k})$ (D) $E_0 q(0.4\hat{i} - 3\hat{j} + 0.8\hat{k})$

PWO218

10. An electron is constrained to move along the y-axis with a speed of 0.1 c (c is the speed of light) in the presence of electromagnetic wave, whose electric field is $\vec{E} = 30\hat{j} \sin(1.5 \times 10^7 t - 5 \times 10^{-2} x) V/m$.
The maximum magnetic force experienced by the electron will be :
(Given $c = 3 \times 10^8$ ms^{-1} and electron charge = 1.6×10^{-19} C)
(A) 1.6×10^{-19} N (B) 4.8×10^{-19} N (C) 3.2×10^{-18} N (D) 2.4×10^{-18} N

PWO219

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	C	C	C	B	194	B	C	B

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Electromagnetic Waves and Wave Optics : RACE-02

1. The wavelength of light coming from a sodium source is 589 nm. What will be its wavelength is water? Refractive index of water = 1.33.

PWO220

2. Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2. After this the phase difference between the two waves is:

(A) $\frac{2\pi}{\lambda}(n_1L_1 - n_2L_2)$ (B) $\frac{2\pi}{\lambda}\left(\frac{L_2}{n_1} - \frac{L_1}{n_2}\right)$ (C) $\frac{2\pi}{\lambda}\left(\frac{L_1}{n_1} - \frac{L_2}{n_2}\right)$ (D) $\frac{2\pi}{\lambda}(n_2L_1 - n_1L_2)$

PWO221

3. The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is _____ *mm*.
[Refractive index of air = 1.0003, wavelength of yellow light in vacuum = 6000 Å]

PWO222

4. In an interference experiment the ratio of amplitudes of coherent waves is $\frac{a_1}{a_2} = \frac{1}{3}$. The ratio of maximum and minimum intensities of fringes will be :

(A) 4 (B) 2 (C) 9 (D) 18

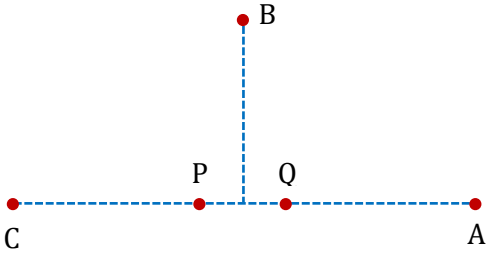
PWO223

5. Interference fringes are observed on a screen by illuminating two thin slits 1 *mm* apart with a light source (λ = 632.8 *nm*). The distance between the screen and the slits is 100 *cm*. If a bright fringe is observed on a screen at a distance of 1.27 *mm* from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close to :

(A) 1.27 μ*m* (B) 2 *nm* (C) 2.87 *nm* (D) 2.05 μ*m*

PWO224

6. In the figure below, *P* and *Q* are two equally intense coherent sources emitting radiation of wavelength 20 *m*. The separation between *P* and *Q* is 5 *m* and the phase of *P* is ahead of that of *Q* by 90°. *A*, *B* and *C* are three distinct points of observation, each equidistant from the midpoint of *PQ*. The intensities of radiation at *A*, *B*, *C* will be in the ratio:



(A) 0 : 1 : 2 (B) 4 : 1 : 0 (C) 0 : 1 : 4 (D) 2 : 1 : 0

PWO225

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7. Two coherent light sources having intensity in the ratio $2x$ produce an interference pattern. The ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be :

(A) $\frac{2\sqrt{2x}}{x+1}$ (B) $\frac{\sqrt{2x}}{2x+1}$ (C) $\frac{\sqrt{2x}}{x+1}$ (D) $\frac{2\sqrt{2x}}{2x+1}$

PWO226

8. The two light beams having intensities I and $9I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\frac{\pi}{2}$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be:

(A) 2 I (B) 6 I (C) 5 I (D) 7 I

PWO227

9. Laser light of wavelength 630 nm incident on a pair of slits produces an interference pattern in which the bright fringes are separated by 8.1 mm. A second light produces an interference pattern in which the fringes are separated by 7.2 mm. Then the wavelength of the second light will be

PWO228

10. Three coherent, equal intensity light rays arrive at a point P on a screen to produce an interference minimum of zero intensity. If any two of the rays are blocked, the intensity of the light at P is I_1. What is the intensity of the light at P if only one of the rays is blocked?

(A) 0 (B) $I_1/2$ (C) I_1 (D) $2I_1$

PWO229

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	443 nm	A	2	A	A	D	D	B	560nm	C

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Electromagnetic Waves and Wave Optics : RACE-03

1. In a Young's double slit experiment, slits are separated by 0.5 *mm*, and the screen is placed 150 *cm* away. A beam of light consisting of two wavelengths, 650 *nm* and 520 *nm*, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is :

(A) 9.75 *mm* (B) 15.6 *mm* (C) 1.56 *mm* (D) 7.8 *mm*

PWO230

2. In a Young's double slit experiment, the slits are placed 0.320 *mm* apart. Light of wavelength λ = 500 *nm* is incident on the slits. The total number of bright fringes that are observed in the angular range $-30° \le \theta \le 30°$ is

(A) 320 (B) 641 (C) 321 (D) 640

PWO231

3. If the source of light used in a Young's double slit experiment is changed from red to violet :

(A) consecutive fringe lines will come closer.

(B) the central bright fringe will become a dark fringe.

(C) the fringes will become brighter.

(D) the intensity of minima will increase.

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PWO232

4. In a Young's double slit experiment, the path different, at a certain point on the screen, between two interfering waves is $\frac{1}{8}th$ of wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is close to :

(A) 0.94 (B) 0.74 (C) 0.85 (D) 0.80

PWO233

5. In a Young's double slit experiment, the separation between the slits is 0.15 *mm*. In the experiment, a source of light of wavelength 589 *nm* is used and the interference pattern is observed on a screen kept 1.5 *m* away. The separation between the successive bright fringes on the screen is:

(A) 6.9 *mm* (B) 5.9 *mm* (C) 4.9 *mm* (D) 3.9 *mm*

PWO234

6. In a double slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}th$ of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is :

(A) 0.568 (B) 0.672 (C) 0.760 (D) 0.853

PWO235

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7. In a Young's double slit experiment, light of 500 *nm* is used to produce an interference pattern. When the distance between the slits is 0.05 *mm*, the angular width (in degree) of the fringes formed on the distance screen is close to :

(A) 0.07° (B) 0.17° (C) 1.7° (D) 0.57°

PWO236

8. In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern.

(A) 1 : 4 (B) 3 : 1 (C) 4 : 1 (D) 2 : 1

PWO237

9. In Young's double slit arrangement, slits are separated by a gap of 0.5 *mm*, and the screen is placed at a distance of 0.5 *m* from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 Å is :-

(A) $1178 \times 10^{-9}\ m$ (B) $1178 \times 10^{-6}\ m$ (C) $1178 \times 10^{-12}\ m$ (D) $5890 \times 10^{-7}\ m$

PWO238

10. In young's double slit experiment performed using a monochromatic light of wavelength λ, when a glass plate ($\mu = 1.5$) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be:

(A) 3 (B) 2 (C) 1.5 (D) 0.5

PWO239

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	A	C	B	D	D	C	B	B

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Electromagnetic Waves and Wave Optics : RACE-04

1. Orange light of wavelength 6000×10^{-10} *m* in illuminates a single slit of width 0.6×10^{-4} *m*. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____.

PWO240

2. A monochromatic light of wavelength 6000Å is incident on the single slit of width 0.01 *mm*. If the diffraction pattern is formed at the focus of the convex lens of focal length 20 *cm*, the linear width of the central maximum is :

(A) 60 *mm* (B) 24 *mm* (C) 120 *mm* (D) 12 *mm*

PWO241

3. A parallel beam of monochromatic light of wavelength 5000 Å is incident normally on a single narrow slit of width 0.001 *mm*. The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction of _____ (degree).

PWO242

4. In a single slit diffraction pattern, a light of wavelength 6000 Å is used. The distance between the first and third minima in the diffraction pattern is found to be 3 *mm* when the screen in placed 50 *cm* away from slits. The width of the slit is _____ $\times 10^{-4}$ *m*.

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PWO243

5. The diffraction pattern of a light of wavelength 400 *nm* diffracting from a slit of width 0.2 *mm* is focused on the focal plane of a convex lens of focal length 100 *cm*. The width of the 1st secondary maxima will be :

(A) 2 *mm* (B) 2 *cm* (C) 0.02 *mm* (D) 0.2 *mm*

PWO244

6. A polarizer - analyser set is adjusted such that the intensity of light coming out of the analyser is just 10% of the original intensity. Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduce the output intensity to be zero, is

(A) 18.4° (B) 71.6° (C) 90° (D) 45°

PWO245

7. A source of light is placed in front of a screen. Intensity of light on the screen is *I*. Two Polaroids P_1 and P_2 are so placed in between the source of light and screen that the intensity of light on screen is $I/2$. P_2 should be rotated by an angle of _____ (degrees) so that the intensity of light on the screen becomes $\frac{3I}{8}$.

PWO246

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8. A light whose electric field vectors are completely removed by using a good Polaroid, allowed to incident on the surface of the prism at Brewster's angle. Choose the most suitable option for the phenomenon related to the prism.

(A) Reflected and refracted rays will be perpendicular to each other

(B) Wave will propagate along the surface of prism

(C) No refraction, and there will be total reflection of light.

(D) No reflection and there will be total transmission of light.

PWO247

9. An unpolarized light beam is incident on a surface at an angle of incidence equal to Brewster's angle. Then:

(A) The reflected and the refracted beams are both partially polarized.

(B) The reflected beam is partially polarized and the refracted beam is completely polarized and are at right angled to each other.

(C) The reflected beam is completely polarized and the refracted beam is partially polarized and are at right angled to each other.

(D) Both the reflected and the refracted beams are completely polarized and are at right angled to each other.

PWO248

10. Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between these polaroids, then the intensity of light emerging from the last polaroid will be :-

(A) $\left(\frac{I_0}{8}\right)\sin^2 2\theta$ (B) $\left(\frac{I_0}{4}\right)\sin^2 2\theta$ (C) $\left(\frac{I_0}{2}\right)\cos^2 \theta$ (D) $I_0 \cos^4 \theta$

PWO249

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	198	B	30.00	2	A	A	30	D	C	A

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